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Studierichting: Informatica
Oefeningenreeks 4A nr. 42.13

$$\int \frac{4x+6}{x^2+4x+3}dx$$

$$\Delta = 4^2 - 4 \cdot 3 = 4 \qquad x_{1,2} = \frac{-4 \pm \sqrt{\Delta}}{2}, \qquad x_1 = -1,\ x_2 = -3$$

$$\int \frac{4x+6}{(x+1)(x+3)}dx = \frac{A}{x+1} + \frac{B}{x+3}$$

Dan is

$$A = \left.\frac{4x+6}{x+3}\right|_{x=-1} = 1$$

$$B = \left.\frac{4x+6}{x+1}\right|_{x=-3} = 3$$

$$\Rightarrow \int \left(\frac{1}{x+1} + \frac{3}{x+3} \right) dx$$

$$= \int \frac{1}{x+1}dx + 3 \int \frac{1}{x+3}dx$$

$$= \ln|x+1| + 3\ln|x+3| + C$$

References